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MIDNIGHT BUILDATHON · WAVE 3

Kola ID

Verify once. Prove anywhere. Stay unlinkable.

Reusable private KYC and credential proofs on Midnight, for the apps that onboard Africa: mobile money, exchanges, lenders, and beyond.

Live on Midnight Preprod | 43 tests passing | Open source · Apache-2.0

Mbanya Dev

github.com/mbanyadev/kola-id · mbanyadev.github.io/kola-id

THE PROBLEM

Every app re-checks the same person, and keeps a copy of their ID

31.0 M

mobile-money accounts in Cameroon at end-2024, 65% of CEMAC's 51.3 M

BEAC, Rapport sur les services de paiement dans la CEMAC 2024, table 13

1.1 bn

registered mobile-money accounts in Sub-Saharan Africa in 2024, up 19%

GSMA, State of the Industry Report on Mobile Money 2025

634,589

mobile-money points of service in CEMAC, up 41% in one year

BEAC 2024

- The same customer passes KYC again at every agent network, exchange and lending app. Each check costs about **US$1 to 1.35** (Sumsub, Blockpass list prices).
- Each app then stores ID scans it must protect under Cameroon's new data-protection law (No. 2024/017).
- Outside finance it's worse: betting sites must keep under-21s out, buyers can't verify co-op members, employers can't verify diplomas.

THE SOLUTION

Check once with a trusted issuer. Prove to any app, privately.

Issuer

Kumba Verify checks Manka once

Signs a credential. Only a **commitment** goes on-chain, into a registry of approved issuers.

→

Holder

Manka's wallet keeps it private

Her birth date, country and screening stay on her device as Midnight private state.

→

App

Sawa Exchange gets a yes

A zero-knowledge proof that she meets **its public policy**, plus a per-app nullifier.

One account per person: the nullifier stops duplicate sign-ups.

Unlinkable: a different nullifier for every app, so apps can't pool data to track her.

Revocable: the issuer can revoke; the proof then fails everywhere.

WHAT MIDNIGHT MAKES POSSIBLE

Public where it must be, private everywhere else

Public on Midnight

- Approved issuer key hashes (who may issue)
- Credential commitments in a Merkle tree, and revocations
- Each app's policy: minimum age, allowed countries, screening freshness, epoch
- Per-app nullifiers and a verification counter (for billing)

Never leaves the holder

- Name, birth date, country, document
- Which issuer checked her
- Which other apps she uses
- Her holder secret

admin	`4c2ac2ed79…1d38c2`
issuers	`e98b7a1e24…b060e4`

credentials	3 issued, 1 revoked · root `57b280917c…345f72`
	active commitments:
	`ad1de8223d…7fade1`
	`0c63871901…9dabf1`

policies	ndole-games: 1 verifications sawa-exchange: 2 verifications
nullifiers	`2a3e6abde9…0ffc94` `875fe3c348…fdb720`

No names, birth dates, countries or issuer-per-person anywhere above.

In Compact, only what passes through `disclose()` becomes public: here the app id, the yes/no eligibility result, the nullifier, Merkle roots, the policy and the date. Never the credential's contents. Right: the demo's public ledger after the full story. Threat model: docs/THREAT_MODEL.md.

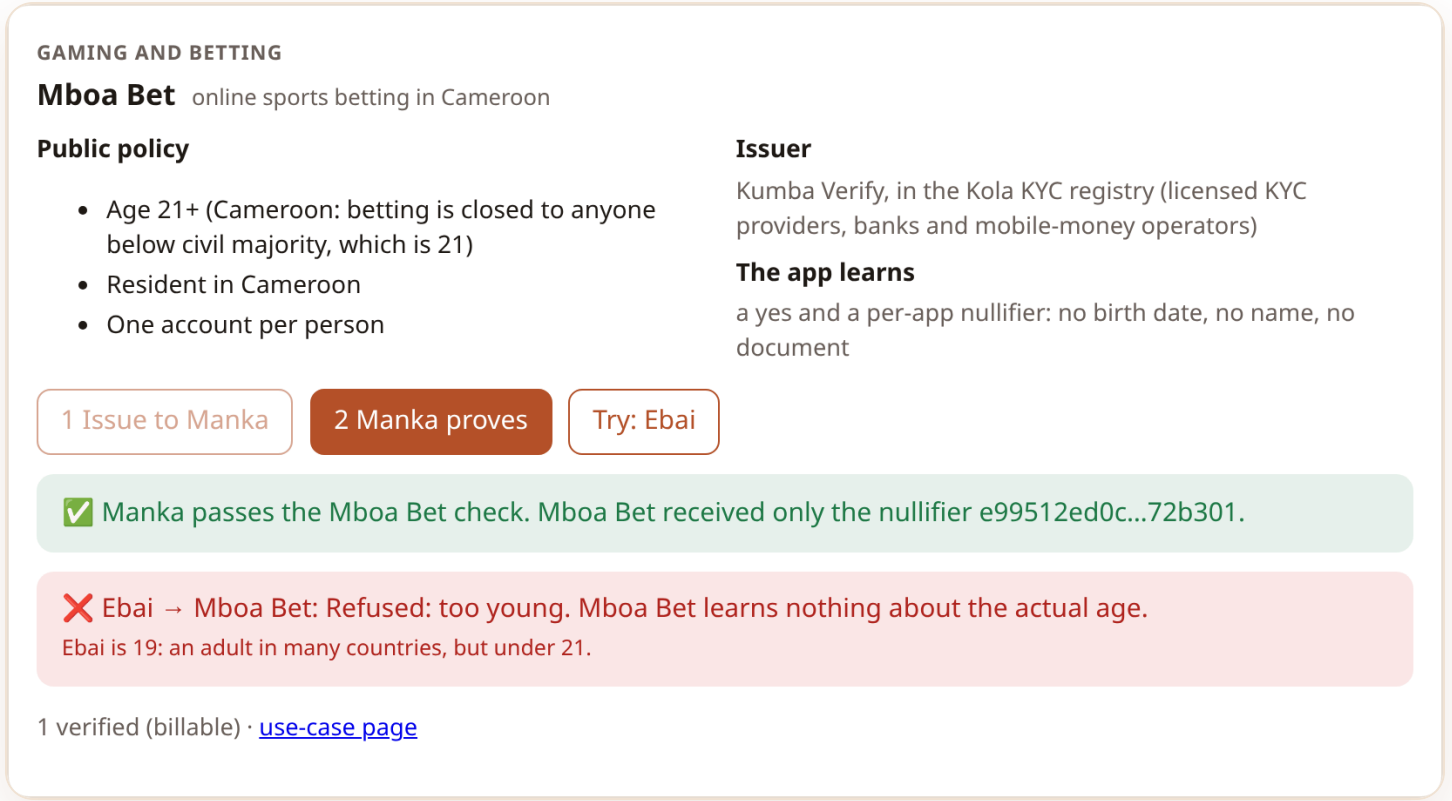
DEMO · MBANYADEV.GITHUB.IO/KOLA-ID

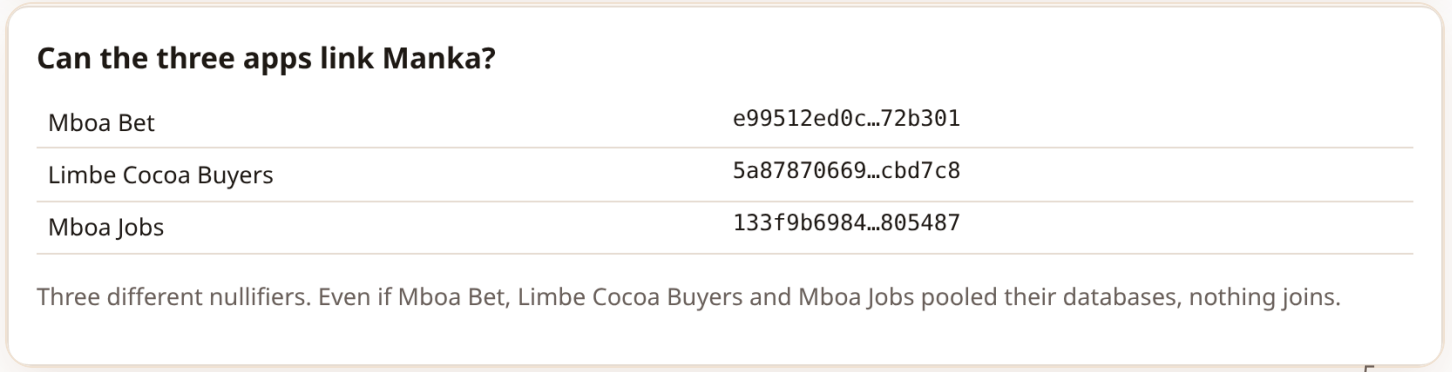
One person, three apps

Manka passes a **betting** app's 21+ check, a **cocoa buyer**'s co-op membership check and a **job platform**'s certificate check, each with a different nullifier.

Refused: Ebai (19) at the betting app, Tabe (lapsed member) at the buyer, a fake diploma mill at issuance.

The real compiled circuits run in the browser. The KYC story is also deployed and replayed on **Preprod**, with every transaction on the explorer.





BEYOND FINANCE · SAME CONTRACT, THREE REGISTRIES

Each trust domain gets its own registry

INDUSTRY	ISSUER → APP	PUBLIC POLICY	WHY IT MATTERS
Gaming	KYC provider → betting operator	21+, Cameroon, one account per person	Betting is closed below civil majority (21) and online operators must prevent minors' access (Decree 2019/2300/PM art. 16; Penal Code art. 249-7)
Agriculture	Cocoa co-op union → buyer	Current member (re-checked ≤ 365 days), adult, Cameroon	EUDR applies from 30 Dec 2026 for large and medium operators; FAO is supporting Cameroon co-ops to comply
Jobs	Accredited institute → job platform	Certificate authenticated ≤ 2 years, not expired, adult	Cameroon's public service launched a diploma-authentication platform in Oct 2024; private employers have none

A co-op union can't mint ID checks and an institute can't mint memberships, because each registry approves only its own issuers. This is enforced by the existing contract and covered by tests. A typed claim field (one registry, many credential types) is on the v2 roadmap.

BUILT ON MIDNIGHT

A real contract, deployed and tested

- **Compact** contract (compiler 0.31.1): addIssuer, removeIssuer, issueCredential, revokeCredential, setPolicy, proveForApp, proveReturning
- Holder data held as **private state** through witnesses; Merkle-tree membership with root history
- **TypeScript SDK** and a simulator running the compiled circuits
- **Midnight.js 4.1.1**, proof server 8.1.3, Lace via the DApp Connector
- **43 Vitest tests**: policy edges, revocation, duplicates, epochs, unlinkability, the on-chain story and the three non-finance registries

Live on Preprod

`d79be6ed9ae73874daa10a1e9e3b36f8772a5d7eb7991f18af1b8975c9027be3`

Deployed 27 Oct 2026. The full story (approve issuer, issue, publish policies, sign-ups, refusals, revocation) was run on-chain, and every transaction is linked to the Midnight explorer in the repo.

The Co-op and Skills registry presets run in the in-browser simulator only; they are not deployed to Preprod yet.

BUSINESS MODEL (PROPOSED)

Apps pay per proof. Issuers earn when their checks are reused.

Pilot

Free

first 1,000 proofs per app, for design partners

Standard

150 FCFA

(≈ €0.23) per sign-up proof, 10 FCFA per returning login. No monthly minimum.

Compliance · roadmap

300 FCFA

per proof, with issuer file access under a reliance agreement

- **40% to the issuer** whose credential was used; 60% to Kola ID. Billing is auditable from the on-chain verification counter.
- Versus a fresh check: US$1.35 per verification on Sumsub Basic (US$149/month minimum) and US$1.00 to 1.35 on Blockpass (US$99/month minimum).
- **Regulated finance:** CEMAC Regulation 02/24 (art. 69 to 72) lets institutions rely on a regulated third party for due diligence, with documents provided on request. Kola ID is the rails for that reliance, not a way around it.

GO-TO-MARKET AND COMPETITION

Start with one issuer and one app per vertical

1. **KYC:** a KYC provider or microfinance institution in Douala or Buea, with one exchange or lending app
2. **Co-ops:** one South-West cocoa union, with one buyer preparing for EUDR
3. **Skills:** one technical institute, with one job platform
4. A hosted check page and webhook, so apps never touch Midnight directly
5. Then mainnet, and the other five CEMAC countries, which share one AML regulation

Proposed plan. No issuers or apps have signed yet.

PLAYER	KOLA ID'S ANGLE
Smile ID, Youverify African KYC leaders	Potential **issuers**: Kola makes their checks reusable without sharing the person
Sumsub ID vendor reusable KYC	Reuse inside one vendor, and apps still get personal data
Privado ID ZK credentials, EVM	Closest concept. Kola uses Midnight private state, per-app nullifiers and on-chain issuer registries, with CEMAC policies
World ID	Proves "unique human" via biometrics, not age, membership or skills
Hyperledger Identus	An SSI toolkit, with no registry, policies or billing

WHERE WE ARE AND WHAT'S NEXT

Early product, clear route to market

Done

- Contract live on Preprod; full story run on-chain
- SDK, in-browser demo, Lace connection
- 43 tests; threat model; three non-finance registries
- Business case with cited market and regulatory sources

Next

- Typed claims: several credential types in one registry
- Compliance mode: an encrypted pointer only the issuer can open, on lawful request
- Hosted check page and issuer SDK; mainnet fee measurements
- First design partners in each vertical

The ask: Midnight Build Club support, and introductions to one issuer and one app in each vertical.

Honest status: no users, customers or revenue yet. All people and companies in the demo are fictional. Sources: docs/BUSINESS.md.